

Lem dim hausdorff con cantor cota superior

Lema 1. Sea $C \subset [0, 1]$ el conjunto de Cantor. Entonces, $\dim_H(C) \leq \frac{\log 2}{\log 3}$.

Demostración: Tomemos $\alpha := \frac{\log 2}{\log 3}$. Para probar que $\dim_H(C) \leq \alpha$, basta con demostrar que $H^{\alpha,*}(C) < \infty$. Para ello, basta con demostrar que $H^{\alpha,*}(C) \leq 1$.

Sea $\delta > 0$. Tomamos $k \in \mathbb{N} : 3^{-k} < \delta$. Entonces, si definimos

$$\forall \ell \in \{0, \dots, 2^k - 1\} : I_\ell^k := \left[\frac{3\ell}{3^k}, \frac{3\ell + 1}{3^k} \right] \subset [0, 1],$$

entonces $C_k = \bigcup_{\ell=0}^{2^k-1} I_\ell^k$ luego $\{I_\ell^k\}_{\ell=0}^{2^k-1}$ es un cubrimiento de C con $\text{diam}(I_\ell^k) = 3^{-k} < \delta$.

$$\therefore H_\delta^{\alpha,*}(C) \leq \sum_{\ell=0}^{2^k-1} \text{diam}(I_\ell^k)^\alpha = 2^k \cdot (3^{-k})^\alpha = 2^k \cdot 3^{-k\alpha} = 2^k \cdot 3^{-k \frac{\log 2}{\log 3}} = 2^k \cdot 2^{-k} = 1.$$

Como $\delta > 0$ era arbitrario, al tomar el límite cuando $\delta \rightarrow 0$, obtenemos que $H^{\alpha,*}(C) \leq 1$. En consecuencia, $\dim_H(C) \leq \alpha = \frac{\log 2}{\log 3}$. ■

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